

Collapse as Open-System Phase-Locking: Basin Dynamics, Born Equilibrium, and the Entropy Bill

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Abstract

Version 3.1 states the collapse claim in its corrected open-system form: measurement is finite-time phase-locking into an attractor basin of a coherence landscape. The measuring device does not pull the wave; it carves the basins — coupling reshapes the joint coherence field $\Lambda = \Lambda_0 \ln C$ so that pointer configurations are its maxima — and the system's own coherence-gradient flow carries the microstate into one basin. Which basin, at equilibrium, is Born-weighted: FRC 100.006's stationary measure integrates over basin j to $p_j \propto |\alpha_j|^2$. The local rise in order during locking is paid on the 566 ledger: the motion of the system-only $Q = S + k^ \ln C$ is the reciprocity current and an export channel must be identified — no identity closes this ledger. Three gates for pillar status are stated with registered routes: norm-controlled dynamics (microstate-distribution flow, avoiding the nonlinear- $|\psi\rangle$ signaling problem), Born-equilibrium*

basin measure, and explicit entropy export. A pilot shows Born frequencies emerging from Langevin flow conditional on the Born-weighted landscape, passing two nulls at $\beta = 4$, with residence times scaling log-linearly in basin depth but not yet scoring as a P2 pass. The v1 vortex/LG material is demoted to a historical appendix; $D \approx 1.90/1.94$ is withdrawn as support.

1. Position in the basin series and lineage

The 100 series carries the current settled statement of each foundational claim. This document is the updated statement of 100.003. It supersedes the v1 vortex/LG paper as the carrier of the collapse claim (v1 retained as Appendix A with corrected caveats) and absorbs the direction of the 100.003.002 working draft with the corrections below. The published v2 was verified (2026-07-07) to remain in the v1 vortex/LG/fractal-dimension direction; v3.1 therefore supersedes both v1 and v2 as the corrected statement, with that lineage retained in Appendix A. Lineage: v1/v2 DOI 10.5281/zenodo.15079820.

Four corrections govern this version:

1. **$D \approx 1.90/1.94$ is withdrawn as support for collapse.** The fractal-dimension values of v1 carry nothing here. The chaos support now comes from [[FRC-100-002]] v2, whose fractal content is the KAM hierarchy and whose ledger results are quantitative (§7).
2. **The ledger identity $\lambda = -d_e S$, present in the 100.003.002 working draft's Gate 3, is removed as an error.** It algebraically forces conservation of the system-only $Q = S + k^* \cdot \ln C$ — the statement [[FRC-566-030]] refutes. In a collapse paper the error is maximally dangerous: Gate 3 exists to demand the entropy bill, and an identity that closes the ledger by definition lets any model pass with bookkeeping instead of physics. The corrected Gate 3 is §5.3.
3. **The collapse dynamics is committed to the microstate-distribution flow** (§5.1), not to a nonlinear deterministic $|\psi\rangle$ evolution, which generically permits superluminal signaling (Gisin-type arguments).
4. **The Born-basin condition is stated with its registered derivation route** (§5.2), not as an unassigned promissory note: the equilibrium half is already published in [[FRC-100-006]]; the open half is named precisely.

2. The corrected thesis

Measurement is finite-time, open-system phase-locking into an attractor basin of a coherence landscape. The device does not attract the wave; it carves the basins. Coupling system to apparatus reshapes the joint coherence field — pointer configurations, the configurations the device can stably resonate with, become the maxima of $\Lambda = \Lambda_0 \ln C$ — and the system's own coherence-gradient dynamics carries the microstate downhill into one basin. The observed outcome is the basin reached. At equilibrium the basin measure is Born: the mouth of each funnel is exactly as wide as $|\alpha_j|^2$. And the funnel has a drain: the local rise in order during locking is paid by entropy export to the surround, on the 566 ledger.

Three consequences of the phrasing, each load-bearing:

A landscape of funnels, not one funnel. The device presents one basin per pointer outcome. Which basin captures a given run is set by the microstate phase flowing under the coherence drift — deterministic flow, practical unpredictability supplied by basin geometry and mixing (§7), not by a postulated random snap.

The funnel is passive; the flow is the system's own. No new force reaches out from the apparatus. The device *tunes* — it reshapes Λ_{eq} through coupling — and the system *locks*, following its own gradient per 787's transport term $\chi \rho \nabla \Lambda$. A funnel guides what falls through it; it does not grab.

Two stages in the double slit, kept distinct. A which-path detector first degrades inter-branch phase coherence — decoherence, the same coherence-channel degradation measured quantitatively in [[FRC-100-002]] v2. Only then does locking select one position basin at the screen. Interference dies at stage one; the definite spot is stage two. Standard decoherence theory does stage one and stops; this paper's added claim is stage two as finite-time relaxation, not a primitive.

Scope of this version. This is a corrected preprint and pilot-supported statement of the 100.003 claim. The three gates of §5 remain open: norm-controlled dynamics, the origin of Born-weighted landscapes, and the explicit entropy export channel.

3. The ledger frame

Per [[FRC-566-030]]'s open-system form, $dS_{\text{total}} + k \cdot d \ln C = \sigma \geq 0$ on locking, with the system-only motion $dQ = \sigma - dS_{\text{env}}$ as the reciprocity current. During locking the local outcome state gains order: S_{local} falls, C_{lock} rises, and $Q = S_{\text{local}} + k \cdot \ln C_{\text{lock}}$ moves — non-conservation of the system-only quantity is the signature of an open exchange, not a defect. Nothing in this paper defines the residual; the requirement is the identification of the export channel that covers the measured motion: apparatus heat, bath entropy, lost phase information, or the measurement record (whose thermodynamic cost is the Landauer/Sagawa–Ueda territory, §6). A model that locks a state but cannot say where the bill went is an attractor story, not an FRC measurement model.

4. The dynamics, at two descriptive levels

Effective/mesoscale (787.787): the coherence field relaxes toward its state-and-coupling-dependent target with finite lag, $\tau_{\Lambda}(\partial_t \Lambda + u \cdot \nabla \Lambda) = D_{\Lambda} \nabla^2 \Lambda - \Lambda + \Lambda_{\text{eq}}[\rho, \psi, E]$. Turning on the measurement coupling changes Λ_{eq} — this is the carving of the basins — and the finite τ_{Λ} is the dynamical origin of finite locking time, hysteresis, and delayed collapse. The microstate density flows as $\partial_t \rho = -\nabla \cdot (\rho v)$, $v \propto \nabla \ln C$ ([[FRC-100-006]]'s continuity form), which is norm-controlled by construction: probability is transported, never manufactured.

Fundamental ([[FRC-100-007]]): Λ as a real scalar field with self-potential and Yukawa coupling, of which the above is the Landau–Ginzburg-style description. This paper operates entirely at the effective level; the fundamental level is cited for completeness, not used.

5. The three gates

5.1 Gate 1 — Norm-controlled dynamics

A schematic non-unitary term added to the Schrödinger equation is not enough, and the obvious repair — a deterministic nonlinear $|\psi\rangle$ flow — walks into the signaling landmine. The committed corpus-native route is the **microstate-distribution flow**: hidden phases ϕ with density $\rho(\phi, t)$, continuity equation, drift up the coherence gradient. The state-vector never evolves nonlinearly; the distribution over microstates relaxes. Embeddings into a stochastic-master-equation or system-plus-bath form are admissible alternatives and are registered for the appendix; any candidate must keep valid states throughout and must be checked explicitly against no-signaling.

5.2 Gate 2 — Born-equilibrium basin measure, with its registered route

For $|\psi_0\rangle = \sum a_n|n\rangle$ the basins must satisfy $\mu(B_n|\psi_0) = |a_n|^2$ at equilibrium — derived, not assigned. The route:

- **Equilibrium half (published, [[FRC-100-006]] Appendix A):** with the coherence functional $C(\phi;\alpha) = \sum_j |\alpha_j|^2 e^{\beta \cos(\phi - \phi_j)}$, the stationary measure of the drift is $\rho \propto \exp[-F/k]$ and its integral over basin j gives $p_j \propto |\alpha_j|^2$ for well-separated basins.
- **Dynamical complement (787 Prediction 2):** basin residence scales as $\exp(\Delta\Lambda/k_{BT_{eff}})$; since $\Lambda = \Lambda_0 \ln C$, the Λ -depth of basin j carries the $\ln|\alpha_j|^2$ offset, so long-run occupancy inherits the Born ratios from the landscape.
- **The open half, named precisely:** *why the landscape carries $|\alpha|^2$ weighting* — i.e., why the coherence functional of the coupled system-apparatus configuration is Born-weighted rather than otherwise. This is the $\Lambda_0 = k^*$ attractor question of the 100.006/Born line. It is registered, open, and shared-credit territory with Bohm–Valentini quantum equilibrium.

The pilot of §9 demonstrates that the route's machinery is genuinely dynamical: capture frequencies emerge from the flow and converge to Born as basins separate, with nulls failing.

5.3 Gate 3 — The entropy bill, correctly stated

During locking, measure $Q_{local}(t) = S_{local} + k^* \cdot \ln C_{lock}$. Its motion is the reciprocity current. The gate: **identify the export channel and show it covers the measured motion** — apparatus heat, bath entropy, record entropy — within an instrumented model where $\sigma \geq 0$ can be verified (the drive-and-bath closure demanded by [[FRC-566-030]] §7, and the dissipation-burst prediction P-T1 of [[FRC-100-005]]). No identity may close this gate; the accounting is physical or it is nothing.

6. Position among neighbors

Every serious neighbor owns a piece of this picture; none owns the connective tissue.

Neighbor	What they own	FRC differential
Zurek (einselection, predictability sieve)	the basin topology: pointer states as survivors of	decoherence stops at the diagonal density matrix; no per-run descent, no

	environmental monitoring	landscape doing work		
Quantum trajectory theory; Mineev et al. 2019	finite-time, continuous, <i>experimentally observed</i> jump dynamics with warning signals — catch-and-reverse	trajectory theory is record-conditioned and stochastic: it describes the locking but supplies no potential, no basin-measure origin, no thermodynamic ledger		
Bohm–Valentini (quantum equilibrium, subquantum H-theorem)	dynamical relaxation to ψ	ψ	ψ ; the equilibrium half of Gate 2, credit shared explicitly	the FRC drift is on a <i>coherence</i> potential — the same object that pays Gate 3 and drives 787 transport
Landauer; Sagawa–Ueda	the bill: entropy production bounds of measurement, feedback, and the record	FRC's claim is that the bill and the basin descent are one ledger, not two literatures		
GRW/CSL; Penrose–Diósi	finite-time objective collapse	by <i>adding</i> noise or gravity; FRC's collapse is relaxation in an existing landscape, no new stochastic ingredient		
Wolynes–Onuchic folding funnels; Haken's synergetics	the mathematical object itself: driven open system relaxing into a basin, concentration paid by dissipation	proof the funnel-with-a-drain is a real object in nature; FRC imports the shape into measurement		

The wager, stated as one falsifiable sentence: $\Lambda = \Lambda_0 \ln C$ is the single object from which the basins (Zurek), the finite-time descent with lead-time markers (trajectory theory/Mineev), the equilibrium measure (Valentini), and the bill (Sagawa–Ueda) all follow. The corresponding kill condition is in §11.

7. The chaos bridge, now with results

[[FRC-100-002]] v2.2 supplies what this paper needs from chaos, as measurements rather than intentions: mixed-phase-space basin geometry with a quantified persistent structure functional; the demonstration that coherence arrests chaos-driven diffusion (localization meeting 787's flight criteria in momentum space); and the coherence-lifetime criterion as an order-one t^{γ_C} crossover. Basin capture in a deterministic mixed system is the classical rehearsal of outcome selection: deterministic flow, practical unpredictability, no structureless randomness required.

8. Predictions

P1 — Finite-time locking with lead time. Collapse events have a continuous early phase preceded by degradation of Λ -stability markers (787 Prediction 3): curvature loss, gradient weakening, variance rise — *before* the transition is visible in the baseline observable. The existing catch-and-reverse dataset (Mineev-class superconducting-qubit experiments) is the out-of-sample arena: FRC's added, losable content is that the markers predict *which* runs jump and *when*, beyond what record-conditioned trajectory theory extracts.

P2 — Residence scaling. Basin dwell times scale exponentially with Λ -depth; occupancy inherits Born ratios from the landscape (787 P2 in the measurement context).

P3 — The bill is concurrent. Entropy export (dissipation bursts, [[FRC-100-005]] P-T1) is synchronized with locking, and its integrated magnitude covers the measured local Q-motion.

P4 — Born from out-of-equilibrium priors. Any sufficiently mixed microstate prior relaxes to Born frequencies under the coherence drift (Valentini-style relaxation, coherence-potential mechanism); deliberately mis-weighted landscapes do not.

9. Pilot status — 2026-07-08

Reproducible script and metrics accompany the preprint (`pilot_collapse.py`, seed fixed). Pilot only; scaffold quantities are labeled as such; wins and wrinkles are reported at equal volume.

Setup. Microstate phase on the circle, Langevin flow $d\phi = \partial_\phi \ln C dt + \sqrt{(2D)} dW$ with $D = k^* = 1$, on 100.006's own functional with three outcomes at Born weights 0.66 / 0.24 / 0.10 and basin gain β . Outcome = basin occupied. **Nothing beyond the flow assigns the basin; capture is the flow's, conditional on the Born-weighted landscape (the Gate-2 open half).**

- **Born capture emerges (Gate-2 machinery, dynamical):** occupancy at $\beta = 4$ is [0.634, 0.247, 0.119] vs targets [0.66, 0.24, 0.10] — max error 2.6%, in the direction and size expected from finite-separation cross-terms. The β -sweep needed to score convergence to the well-separated-basin limit is registered for the appendix; it is not claimed in this release package.
- **Null A (flat landscape, $\beta = 0$):** occupancy $\rightarrow 1/3$ each; fails Born (max deviation 0.33).
- **Null B (equal-weight landscape, unequal Born targets):** occupancy $\rightarrow 1/3$ each; fails Born. Together with the capture result: the frequencies come from the landscape, not the harness.
- **Residence scaling — trend present, NOT scored as a P2 pass:** mean dwell is monotone in basin depth and log-linear (dwell 0.471 / 0.250 / 0.137 for depths $\ln 0.66$ / $\ln 0.24$ / $\ln 0.10$), but the fitted slope is 0.65 against the naive Kramers expectation $1/D = 1$. Two identified estimator problems, reported at full volume: sector-boundary recrossings contaminate short dwells, and per-basin saddle heights and curvature prefactors are not equal. A committor/milestoning dwell definition is registered for the appendix; P2 is scored only after that hardening.
- **Locking ledger (scaffold):** driving β from 4 $\rightarrow$ 12, phase entropy falls 1.530 $\rightarrow$ 1.083 nats while local order rises (C_{lock} 0.886 $\rightarrow$ 0.954); $Q = S + \ln C_{\text{lock}}$ moves by -0.37 nats — **the open-system signature of 566.030: the system-only quantity is not conserved through locking.** Wrinkle at full volume: the "required export" accumulator inherited from the 100.003.002 scaffold mis-handles the thermostatted ramp, where ΔQ is negative and the bill is carried by the Langevin bath; its sign convention is recorded as a scaffold defect. The correct Gate-3 instrument is an instrumented bath in which σ and dS_{env} are separately measured; registered for the appendix.
- **Limit stated in advance and confirmed in operation:** the $|\alpha_j|^2$ weights enter through the coherence functional. The pilot shows the landscape $\rightarrow$ frequency

machinery is dynamical and Born-consistent; it does not derive why the landscape is Born-weighted. That is Gate 2's open half, exactly as named in §5.2.

10. Controls and appendix program

β and D sweeps of the capture error; many-outcome scaling; committor-based dwell estimation before P2 scoring; instrumented-thermostat entropy accounting ($\sigma \geq 0$ verified, export channel itemized) for Gate 3; embedding of the distribution flow in an SME/Lindblad form with an explicit no-signaling check for Gate 1; a family of wrong-landscape nulls; and the Mineev-class dataset reading for P1 — with the lead-time markers pre-registered before any experimental record is examined.

11. Kill conditions

This collapse mechanism fails if: no norm-controlled dynamics consistent with no-signaling can be written; the Born measure cannot be obtained without hand-assignment (including failure of the §5.2 route's open half); the export channel cannot be identified in an instrumented closure, or $\sigma < 0$ is required; finite-time collapse signatures with lead-time structure are excluded by experiment; **or the connective-tissue kill: trajectory theory plus information thermodynamics reproduce everything — including the lead-time statistics of catch-and-reverse experiments — with no residual Λ -structure.** Failure here kills this mechanism. It does not kill 566, whose kill conditions live in the 566 papers, and it does not kill the chaos results of 100.002 v2.

Appendix A — v1, demoted

The 2025 vortex/LG paper (v1/v2, DOI 10.5281/zenodo.15079820) is retained as historical context: vortex attractor dynamics as the first draft of the basin picture, the LG-beam platform as a still-interesting downstream experimental idea. Withdrawn as support: $D = 1.94 \pm 0.05$ and its consistency with the stadium $D \approx 1.90$ (see [[FRC-100-002]] v2 Appendix A); the claim that resonant modes are selected *above* Born probability (v3 targets Born exactly, per Gate 2); and the unconstrained non-unitary dynamics. The LG proposals may be revived only under the Gate 1–3 discipline of this version.

Connections

- [[FRC-100-002]] — Chaos bridge: mixed-phase-space basin geometry and coherence-arrested diffusion

- [[FRC-100-005]] — Thermodynamics of locking; reference Gate-3 instrument and P-T1
- [[FRC-100-006]] — Born-equilibrium basin measure (equilibrium half of Gate 2)
- [[FRC-100-007]] — Lambda-field; outcome statistics live at distribution level per the v3.1 audit
- [[FRC-566-001]] — Entropy-coherence reciprocity and UCC
- [[FRC-566-030]] — Open-system ledger; system-only Q-motion as reciprocity current
- [[FRC-787-787]] — Lambda-flight transport and residence scaling

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